



AI & GenAI Career Roadmap: From Fresher to Expert

A comprehensive guide to building your career in Artificial Intelligence and Generative AI in India, from entry-level positions to senior roles over an 8-year journey. This roadmap covers essential skills, salary expectations, project ideas, and specialized paths to help you navigate this rapidly evolving field.

Your 8-Year AI Career Journey



This progressive journey builds your expertise from foundational skills to leadership roles, with each phase building on the previous one.

Phase 0: Foundations (0-6 months)

Core Programming

- Python fundamentals
- Git version control
- Linux basics

Data Handling

- NumPy & pandas
- SQL databases
- Statistics fundamentals

Visualization & ML

- Matplotlib/Plotly
- Power BI/Tableau
- Excel for data analysis
- scikit-learn basics

MLOps Introduction

- virtualenv/conda
- FastAPI basics
- Docker introduction

Output: 3 mini-projects + 1 portfolio README + ATS resume

Project ideas: Sales dashboard; churn prediction API; real-time CSV → dashboard

Phase 1: Job-ready DA/ML (6-12 months)

Advanced ML Techniques

- Feature engineering
- Cross-validation strategies
- XGBoost/LightGBM
- Class imbalance handling
- Time-series basics

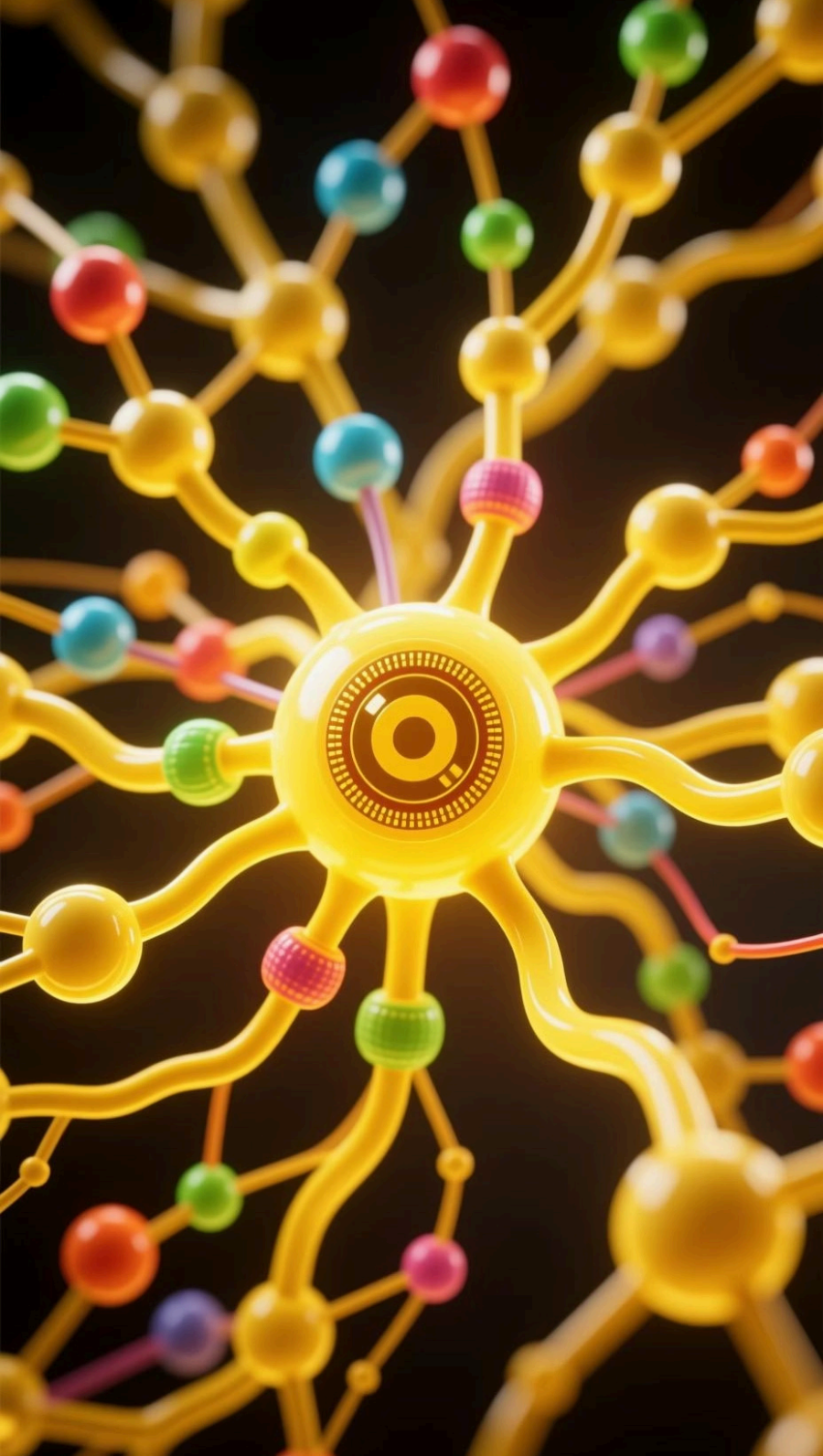
Data Engineering & Deployment

- ETL with Pandas/Polars
- Prefect/Airflow introduction
- FastAPI + Docker deployment
- Render/EC2 hosting
- Monitoring basics (Evidently)

Output: 2 end-to-end case studies with deployment links

Project ideas: Demand forecasting; lead scoring; anomaly detection





Phase 2: Applied DL & NLP (1-2 years)

Deep Learning Foundations

- PyTorch/TensorFlow
- CNN/RNN basics
- Transformers architecture
- Embeddings

Vector Databases

- pgvector
- Qdrant
- Milvus

GenAI Introduction

- Prompt engineering
- Semantic search
- Basic RAG implementation
- Evaluation techniques

Output: 1 NLP + 1 Computer Vision project deployed; CI/CD basics

Project ideas: Resume parser + JD-match; invoice OCR + QA; review sentiment analysis

Phase 3: MLOps → LLMOps (2-3 years)

MLOps Fundamentals

- Package management
- Experiment tracking (MLflow/W&B)
- Data versioning (DVC)
- Scalable APIs

Output: 1 production RAG + monitoring dashboard + postmortem

Project ideas: Document QA; recruiter assistant; analytics copilot

GenAI Advanced Techniques

- RAG patterns and guardrails
- LangChain/LlamaIndex
- vLLM/Transformers inference





Phase 4: Senior AI / GenAI Engineer (3-5 years)



Specialization

Choose a focus area (NLP/GenAI, Vision, RecSys) and own the entire lifecycle from PoC to production



Advanced Techniques

Master LoRA/QLoRA finetuning, task-specific evaluations, LLM judges, tool-use, caching, and multi-tenancy



LLMOps

Implement tracing, evaluation frameworks, safety measures, cost governance, and canary rollouts

Output: 2-3 production systems + architecture docs + cost/latency benchmarks



Phase 5: Staff/Lead (5-8 years)

Technical Leadership

- Multi-agent systems
- Retrieval pipelines at scale
- A/B testing frameworks
- Data contracts
- Service Level Objectives (SLOs)

Organizational Impact

- Roadmap influence
- Mentoring junior engineers
- Vendor/model selection
- Budget ownership
- Measurable business impact

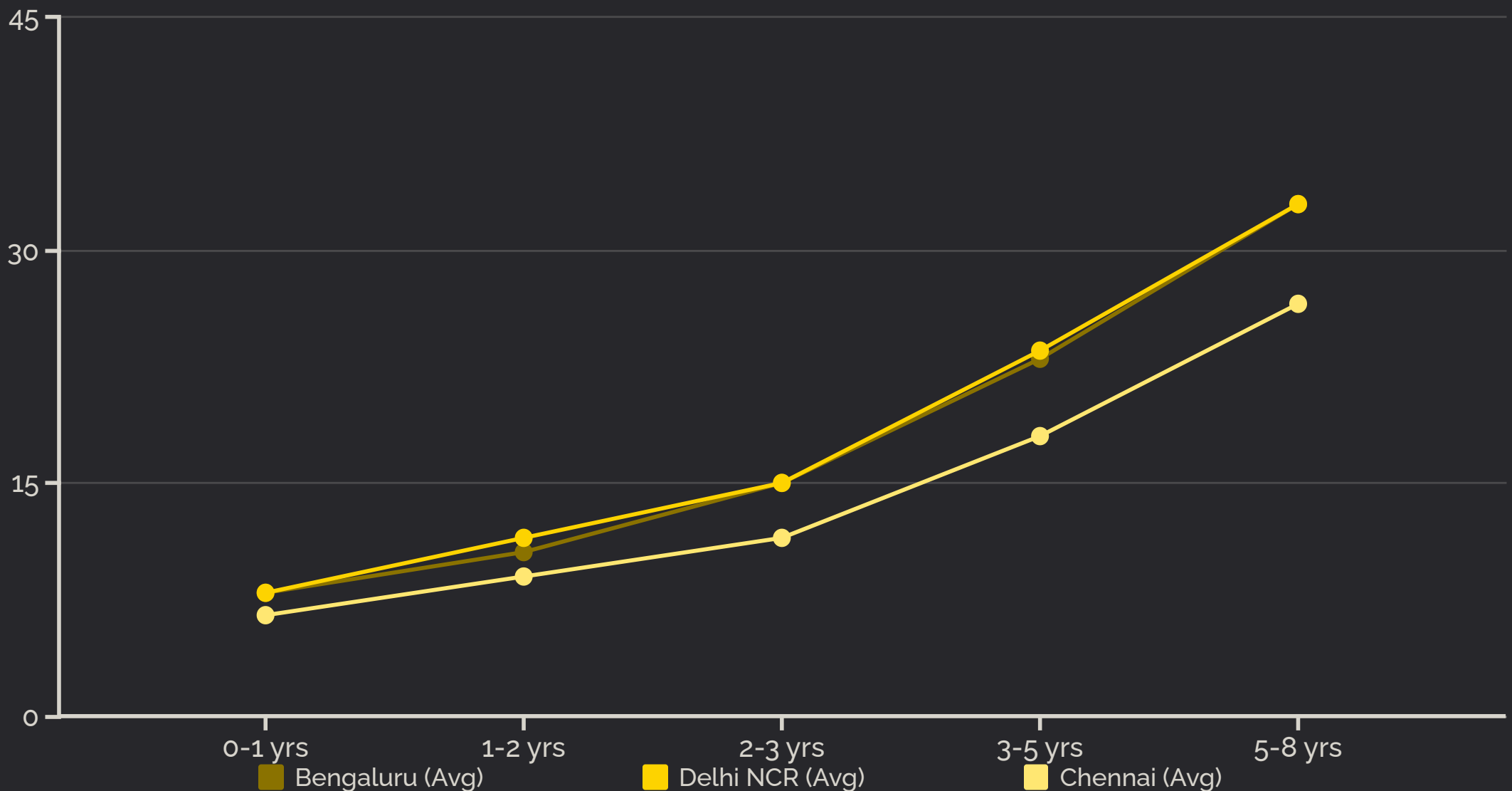
Outputs: Design docs, RFCs, playbooks; measurable business impact

City-wise India Salaries (LPA)

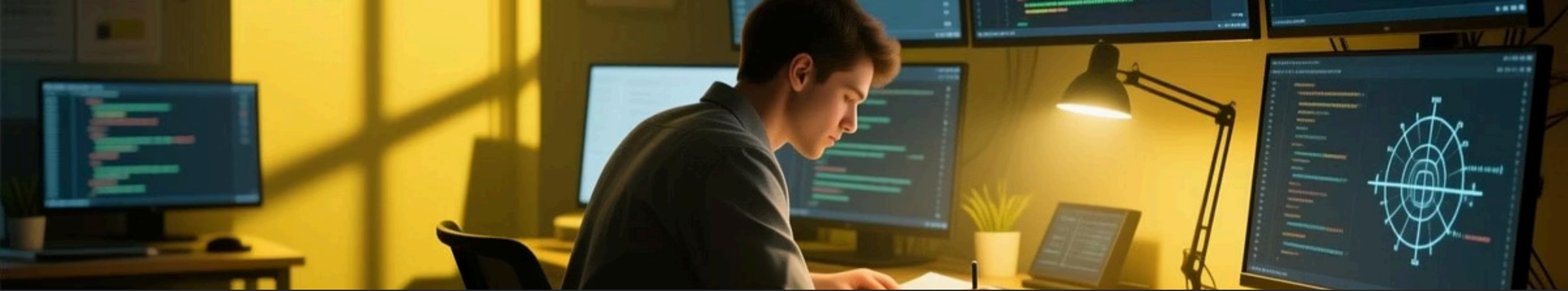
Salary ranges are indicative; GenAI/LLMOps roles can trend 10–30% higher when owning production metrics.

City	0–1 yrs	1–2 yrs	2–3 yrs	3–5 yrs	5–8 yrs
Bengaluru	6–10 L	8–13 L	12–18 L	18–28 L	26–40 L
Hyderabad	5–9 L	8–13 L	11–17 L	17–27 L	25–38 L
Delhi NCR	6–10 L	9–14 L	12–18 L	18–29 L	26–40 L
Mumbai	5–9 L	8–12 L	10–16 L	16–25 L	24–36 L
Pune	5–9 L	8–12 L	10–15 L	15–24 L	22–34 L
Chennai	5–8 L	7–11 L	9–14 L	14–22 L	21–32 L

Salary Growth Trajectory



The chart shows the average salary progression (in LPA) across three major tech hubs in India. Note the significant jump between the 2-3 year and 3-5 year experience levels, highlighting the value of specialized skills and leadership capabilities.



Essential Skills to Master



Math & Statistics

Probability, distributions, confidence intervals, A/B testing, bias/variance, regularization, time-series analysis



Programming

Python, testing (pytest), packaging/OOP; SQL (window functions); Git version control



Data Engineering

pandas/Polars; SQL performance optimization; data cleaning; feature stores (intro)



Machine Learning

scikit-learn; ensemble methods; ML pipelines; SHAP/interpretability; hyperparameter tuning

Advanced Skills for Growth



Deep Learning

PyTorch; CNNs; attention/transformers; transfer learning techniques



NLP

Tokenization; embeddings; vector search; RAG & evaluation methods



Computer Vision

Classification/detection/segmentation; OCR + post-processing



MLOps/LLMOps

FastAPI; Docker; CI/CD; MLflow/W&B; monitoring; retraining; tracing



GenAI

Prompts; function calling; agents; LoRA/QLoRA; vLLM/TensorRT-LLM; cost/latency tuning



Cloud

AWS/GCP/Azure basics; IAM; managed ML services



Security/Compliance

PII redaction; dataset governance; safety evaluations

Project Ideas by Domain

Data Analytics / BI

- Admissions funnel analysis
- Cohort retention dashboard
- Revenue forecasting dashboard

Machine Learning

- Lead scoring with uplift modeling
- Pricing elasticity model
- Credit risk assessment (lite version)

NLP/GenAI

- PDF RAG copilot
- Resume→JD match & rewrite
- Policy Q&A chatbot

Computer Vision

- Invoice/ID OCR + reconciliation
- Defect detection from images

Agentic AI

- Admissions Agent (FAQs+demos+CRM)
- Job-Apply Agent (search/filter/drafts/logs)



Pivot after ~3 Years:
Specializing in GenAI

AI Engineer → GenAI / Agentic AI Engineer

LLM Fundamentals

- Tokenizers, attention mechanisms
- KV-cache optimization
- Quantization techniques
- Batching strategies
- Efficient model serving

Advanced RAG

- Retrieval fusion (BM25+dense)
- Dynamic chunking strategies
- Query rewriting techniques
- Reranking methodologies

This specialized path focuses on the cutting-edge of AI development, with emphasis on generative models and autonomous agent systems.

GenAI Specialization: Advanced Skills

$f(x)$

Function Calling & Tools

Structured outputs, guardrails, authentication systems



Finetuning

LoRA/QLoRA on domain data; synthetic data generation; evaluation metrics (faithfulness, relevancy)



LLMOps at Scale

Tracing, golden test sets, canary deployments, versioning, observability, cost governance



Multi-agent Systems

Planner/worker architectures, state machines, memory systems, retry mechanisms

Security & Compliance

PII handling, content redaction, access controls, prompt-injection defenses



12-Week GenAI Upskill Track

1

Weeks 1-3: RAG Deep Dive

- Hybrid retrieval systems
- Reranking methodologies
- Domain-specific Q&A
- Evaluation suite development

2

Weeks 4-6: Function Calling & Tools

- Structured tool integration
- Implementing guardrails
- Golden-set testing

3

Weeks 7-9: Model Finetuning

- LoRA/QLoRA techniques
- Small model optimization
- RAG vs. finetuning comparisons
- Caching strategies

4

Weeks 10-12: Multi-agent Systems

- Agent architecture design
- GPU/CPU deployment
- Tracing implementation
- Monitoring dashboards
- Cost budget optimization



12-Week Study Plan for Beginners

Week	Focus	Goals	Output
1	Python/SQL refresh + Git	Data handling & query fluency	Mini ETL + SQL challenge
2	Stats & EDA	Hypothesis tests & visualization	EDA notebook on public dataset
3	ML basics	Train/validate models	Baselines + metrics report
4	Feature eng + CV	Improve signal	Feature importance & model compare
5	XGBoost/LightGBM	Tune + avoid leakage	Tuned model & checklist
6	FastAPI + Docker	Serve models	Inference API link

12-Week Study Plan (continued)

Week	Focus	Goals	Output
7	DL intro (PyTorch)	Build & train neural networks	Simple CNN on images or tabular MLP
8	Transformers/NLP	Embeddings & RAG intro	Semantic search demo
9	Vector DB + RAG	Hybrid retrieval	RAG MVP with citations
10	LLMOps evals	Golden sets & tracing	Eval dashboard + logs
11	Finetuning (LoRA)	Domain adaptation	Fine-tuned small model
12	Multi-agent + hardening	Reliability & cost	Prod-ready demo + README

This structured learning plan provides a week-by-week guide to building essential AI skills, with concrete outputs to demonstrate your progress and build your portfolio.

Portfolio Checklist

4–6 Diverse Projects

Include projects across DA/ML/NLP/CV/GenAI domains, with at least 2 deployed demos that showcase your range of skills.

Professional Documentation

Create clean READMEs with problem statements, data descriptions, methodologies, metrics, key decisions, and next steps.

Optimized Resume & LinkedIn

Develop a one-page ATS-friendly resume and enhance your LinkedIn with an effective headline and featured section.

Business Impact Case Studies

Document projects with clear business impact metrics and thoughtful discussion of trade-offs.

Production-Ready Code

Include unit tests for utilities, pre-commit hooks, and basic CI/CD pipelines.

Visual Demonstrations

Add screenshots, GIFs, and short Loom videos to showcase your work in action.

Portfolio Evaluation Rubric



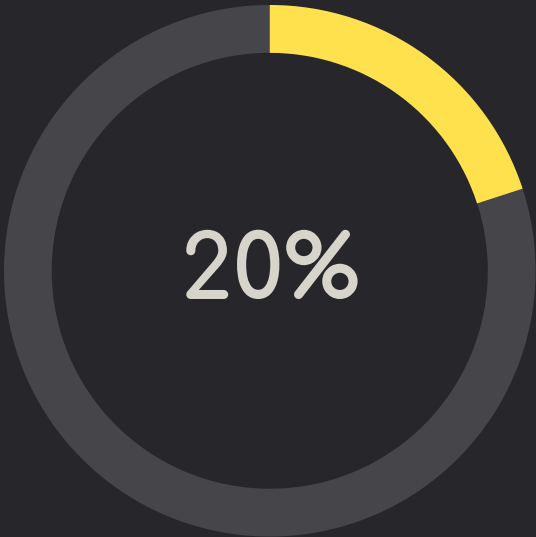
Relevance to Target Roles

Projects align with the specific skills and technologies sought by employers in your target roles



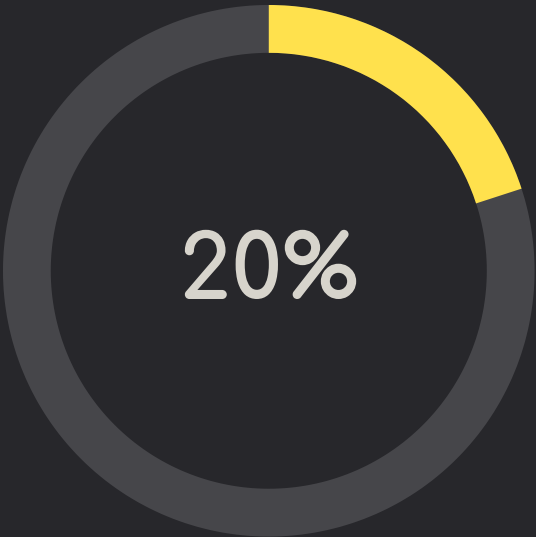
Technical Depth & Correctness

Demonstrates thorough understanding of concepts and implements solutions correctly



Production Readiness

Includes APIs, Docker containers, and monitoring solutions that show operational thinking



Communication

README clarity and storytelling that effectively explains complex concepts



Business Impact

Demonstrates metrics, cost/latency considerations, and A/B testing learnings

Use this rubric to evaluate your portfolio before applying for positions. A strong portfolio that scores well across all five categories will significantly increase your chances of landing interviews and offers in the competitive AI job market.